

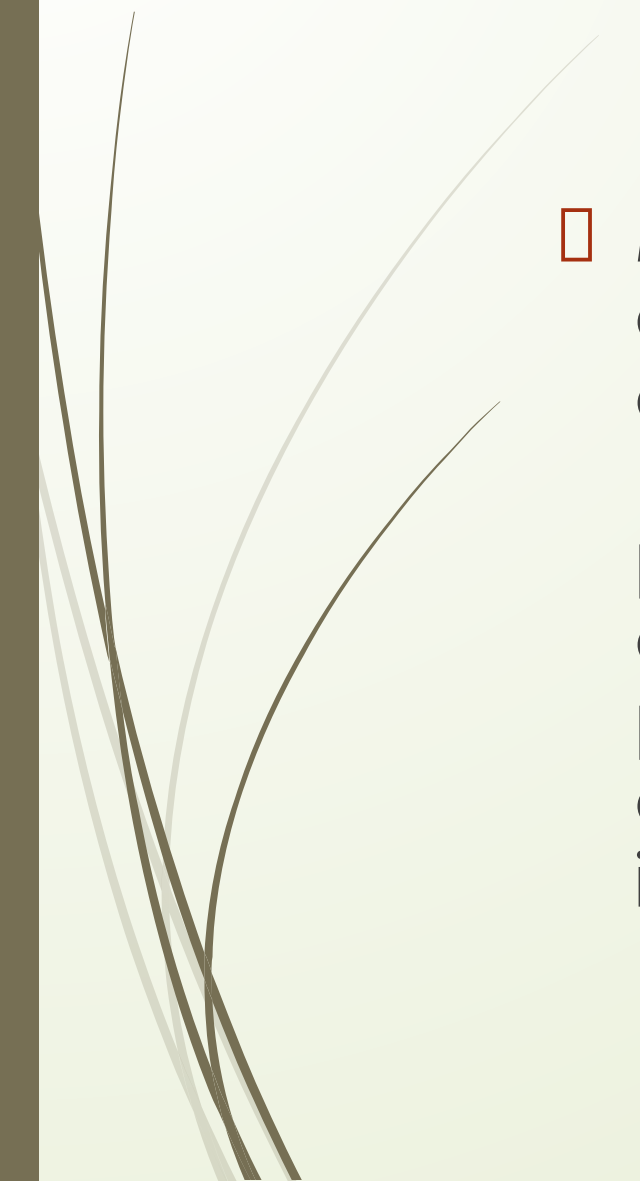


Introduction to Enterprise System and Management

ADV01 – Enterprise System

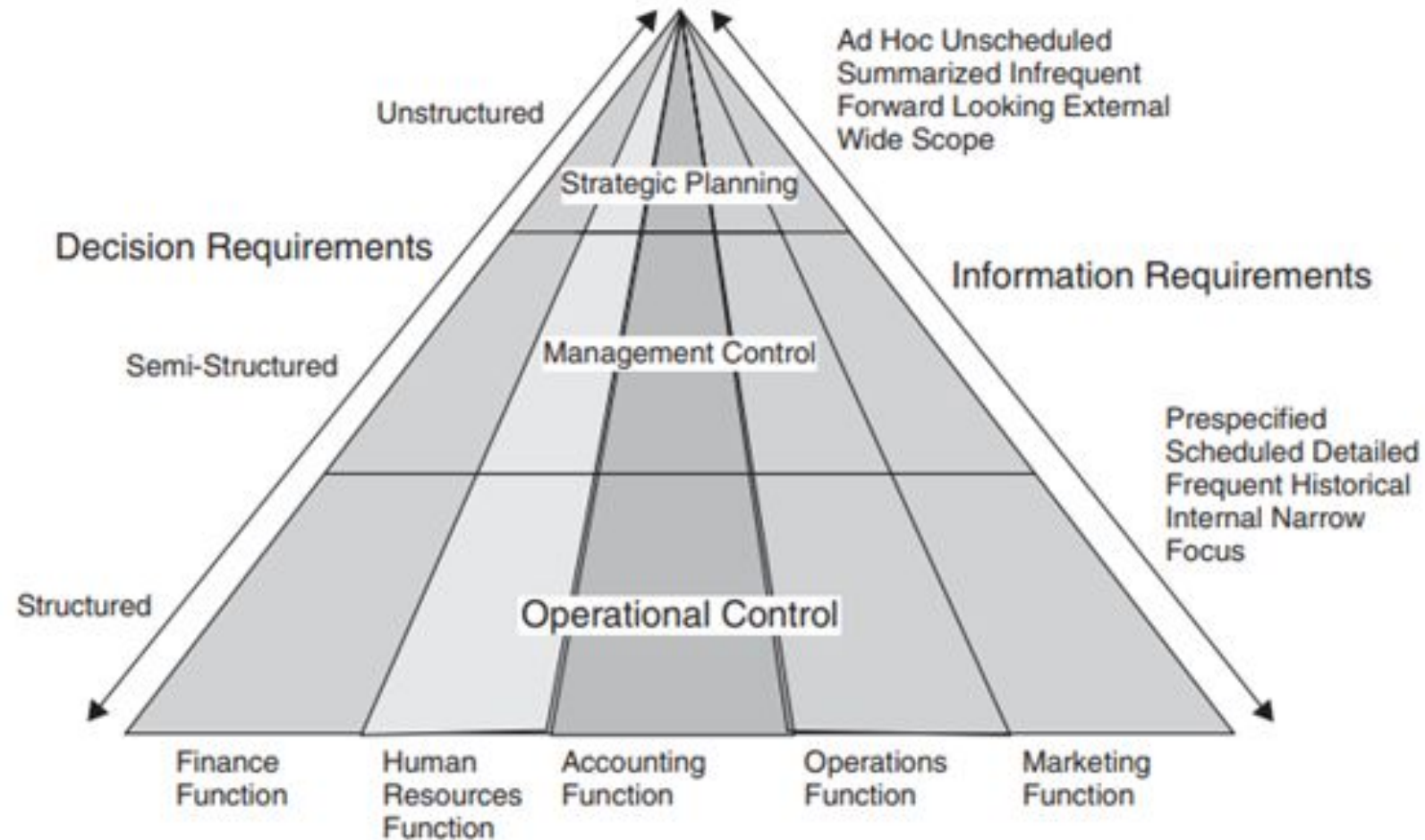


Introduction



- Management did not appreciate the scale of an organization's challenges before, during, and after installing ERP systems in the early days of ERP adoption. ERP systems differ from traditional packaged software such as Microsoft Office and others, despite the fact that they are both packaged software. An ERP implementation entails more than just infrastructure and incompatibilities.

Enterprise System and Management



Management Pyramid with Information Requirements
(Enterprise System, L. Motiwalla)

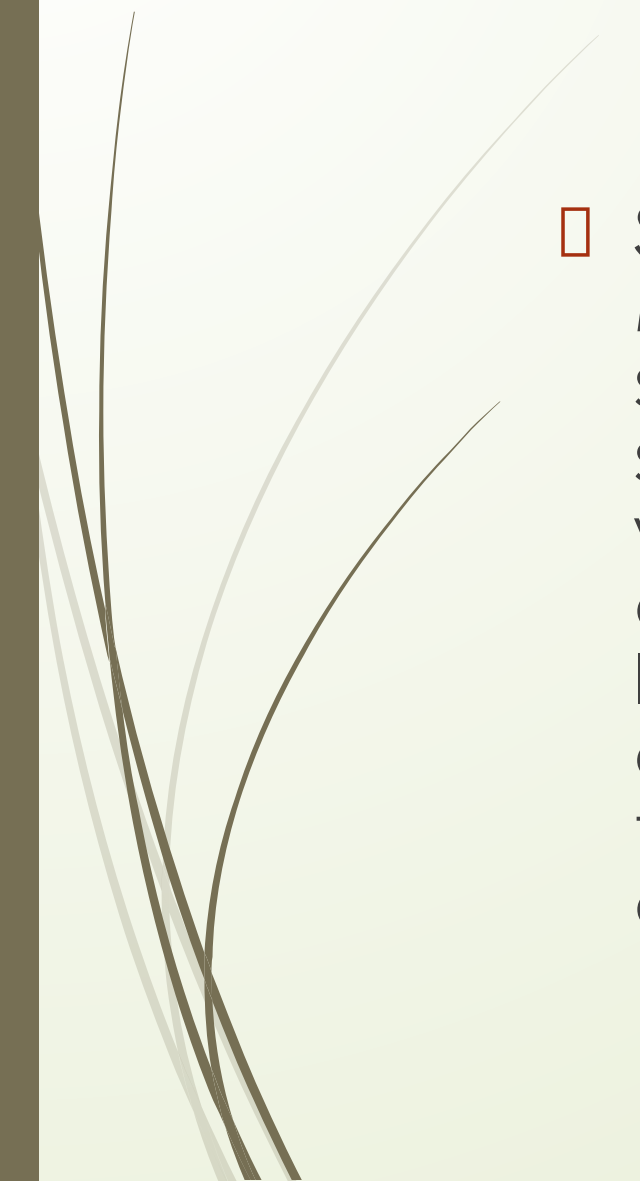


Levels of Management

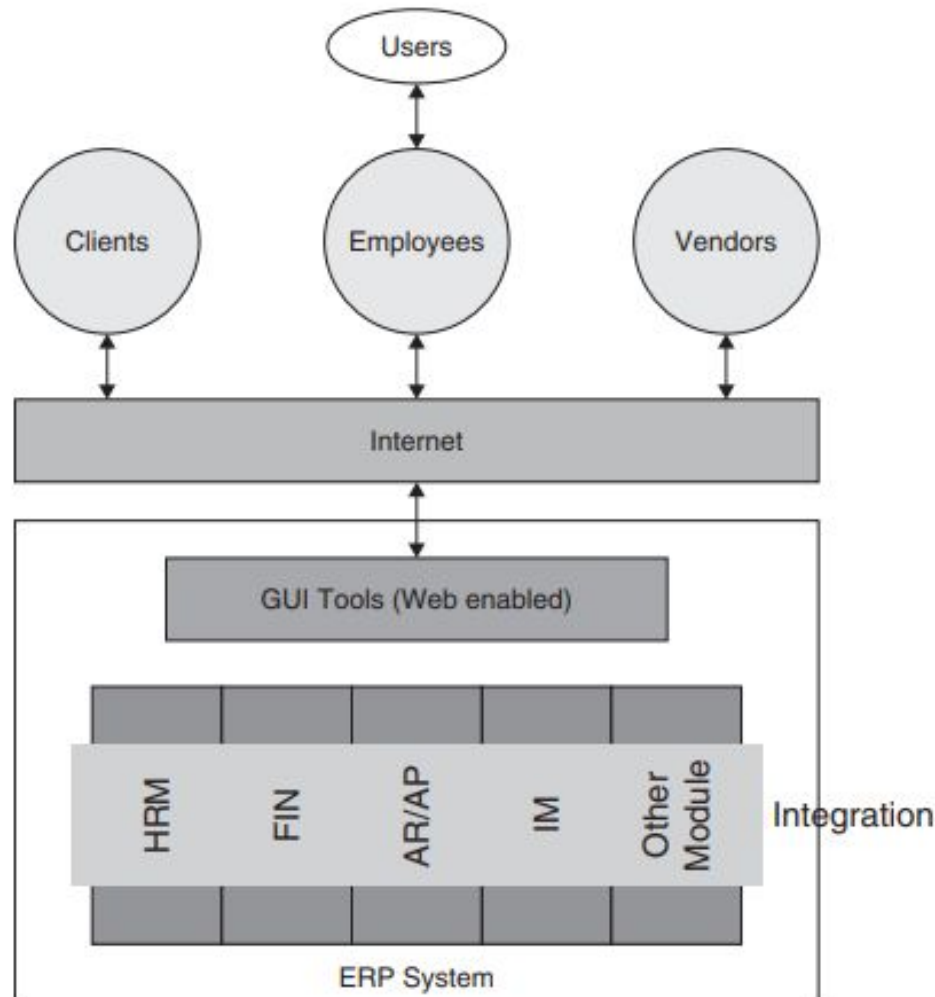
- According to sprigghr.com, there are three levels of Management which are as follows.
 - Administrative, Managerial, or Top Level of Management - consists of an organization's board of directors and the chief executive or managing director. It is the ultimate source of power and authority since it oversees a company's goals, policies, and procedures.
 - Executive or Middle Level of Management - The branch and departmental managers form this middle management level.



Levels of Management (conti)

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- Supervisory, Operative, or Lower Level of Management - This management level consists of supervisors, foremen, section officers, superintendents, and all other executives whose work must do largely with HR oversight and the direction of operative employees. Managers at the lower level are primarily concerned with the execution and coordination of day-to-day workflow that ensures the completion of projects and that deliverables are met.

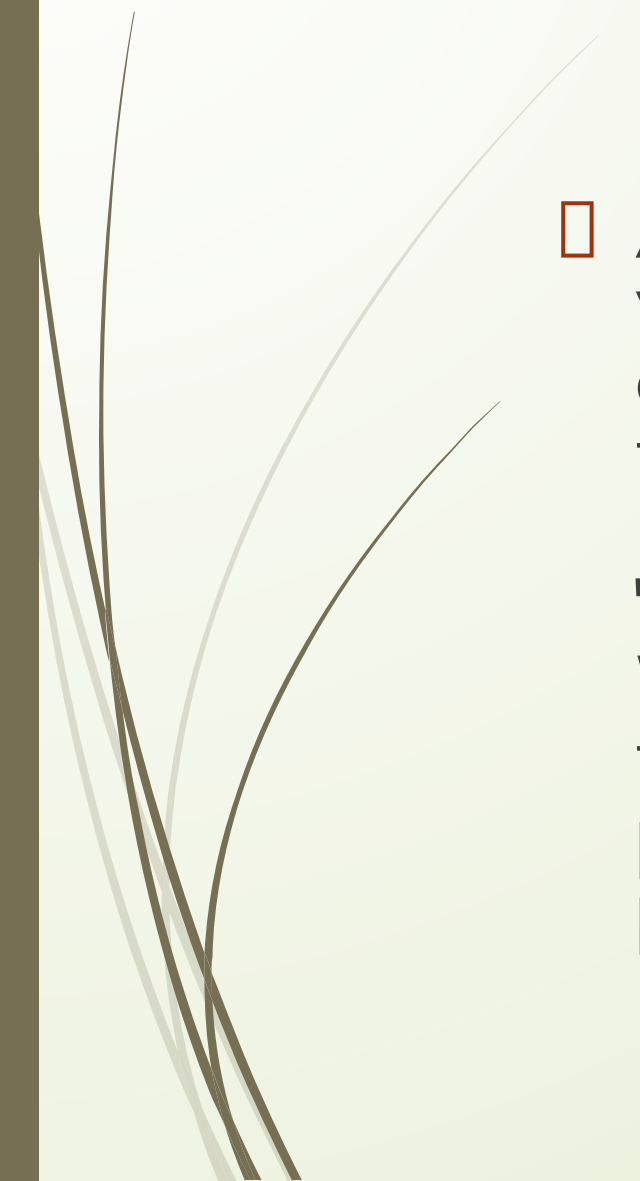
Enterprise Resource Planning



- Enterprise resource planning (ERP) systems are the specific kind of enterprise systems to integrate data across and be comprehensive in supporting all the organization's major functions. ERP systems are comprehensive software applications that support critical organizational functions.

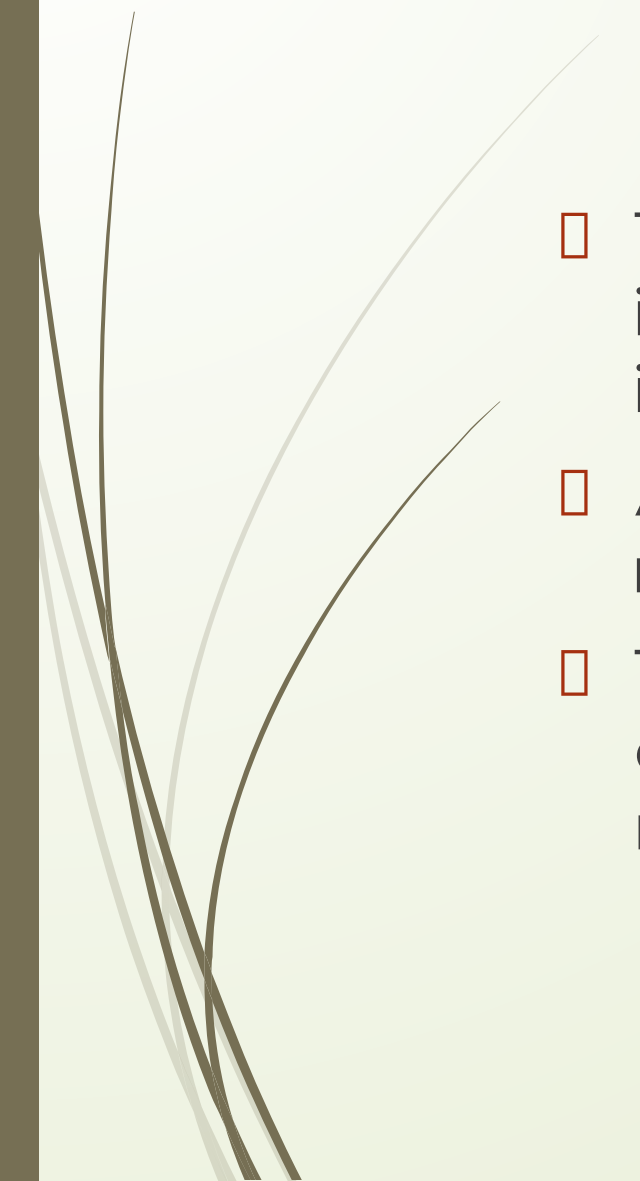


Enterprise Resource Planning

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- As shown in Figure 1-2, they integrate both the various functional aspects of the organization and the systems within the organization with those of its partners and suppliers. Furthermore, these systems are "Web-enabled," meaning that they work using Web clients, making them accessible to all of the organization's employees, clients, partners, and vendors from any time and place.



ERP Goals



- To make information flow be both dynamic and immediate, increasing the usefulness and value of the information.
- Acts as a central repository eliminating data redundancy and adding flexibility.
- To integrate departments and functions across an organization onto a single infrastructure that serves the needs of each department.

Evolution of ERP

Timeline	System	Platform	Description
1960's	Inventory management and control	Mainframe legacy using third-generation Software (e.g., Cobol, Fortran)	With a focus on efficiency, these systems were designed to manage and track inventory of raw materials and guide plant supervisors on purchase orders, alerts, and targets, providing replenishment techniques and options, inventory reconciliation, and inventory reports.
1970's	Material requirements planning (MRP)	Mainframe legacy using third-generation Software (e.g., Cobol, Fortran)	With a focus on sales and marketing, these systems were designed for job shop scheduling

1980's

Manufacturing
requirements planning
(MRP II)

Mainframe legacy using
fourth-generation
database software and
manufacturing
applications.

With a focus on
manufacturing strategy
and quality control,
these systems were
designed for helping
production managers in
designing production
supply chain
processes—from
product planning, parts
purchasing, inventory
control, and overhead
cost management to
product distribution.

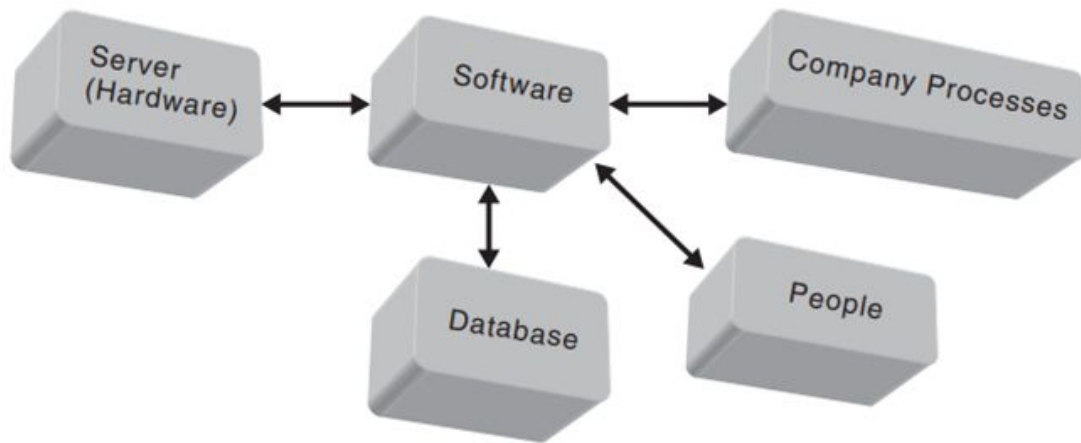
0's	Enterprise resource planning (ERP)	Mainframe or client-server using fourth-generation database software and package software application to support most organizational functions	With a focus on application integration and customer service, these systems were designed for improving the performance of the internal business processes across the complete value chain of the organization. They integrate primary business activities like product planning, purchasing, logistics control, distribution, fulfillment, and sales; additionally, they integrate secondary or support activities like marketing, finance, accounting, and human resources.
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Evolution of ERP (conti)

2000's	Extended ERP or ERP II	Client-server using Web platform, open-source and integrated with fifth-generation applications like SCM, CRM, SFA.	With a focus on agility and a customer-centric global environment, these systems extended the first-generation ERP
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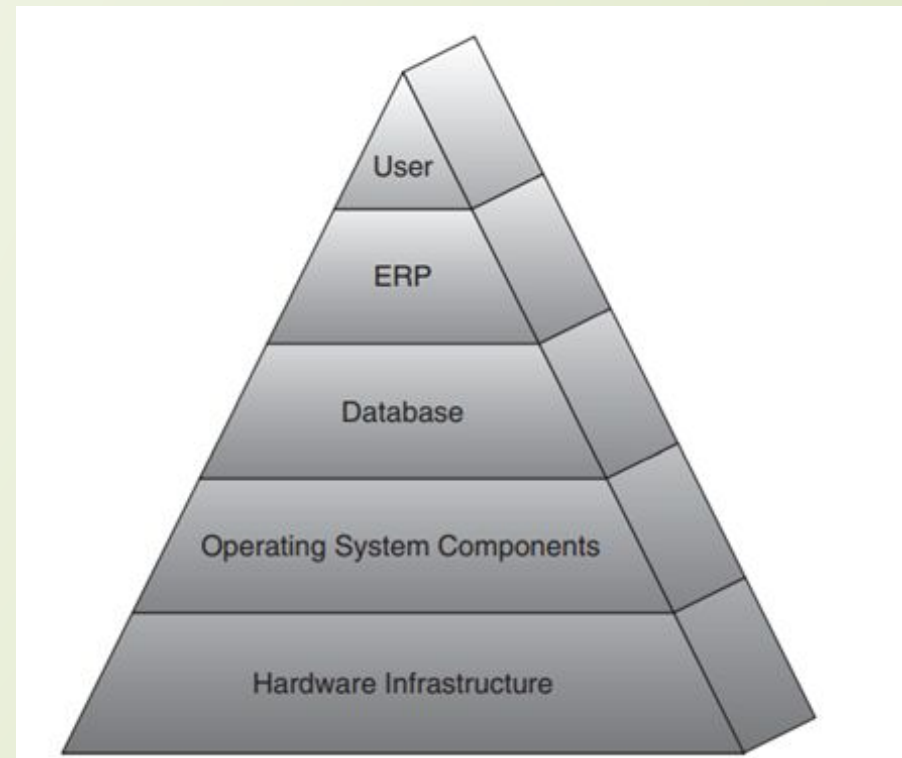
ERP System Components

- An ERP system depends on hardware (i.e., servers and peripherals), Software (i.e., operating systems and database), information (i.e., organizational data from internal and external resources), process (i.e., business processes, procedures, and policies), and people (i.e., end-users and IT staff) to perform the input, process, and output phases of a system. The basic goal of ERP, like any other information system, is to serve the organization by converting data into useful information for all the organizational stakeholders. Figure 1.3 shows the components of an ERP.

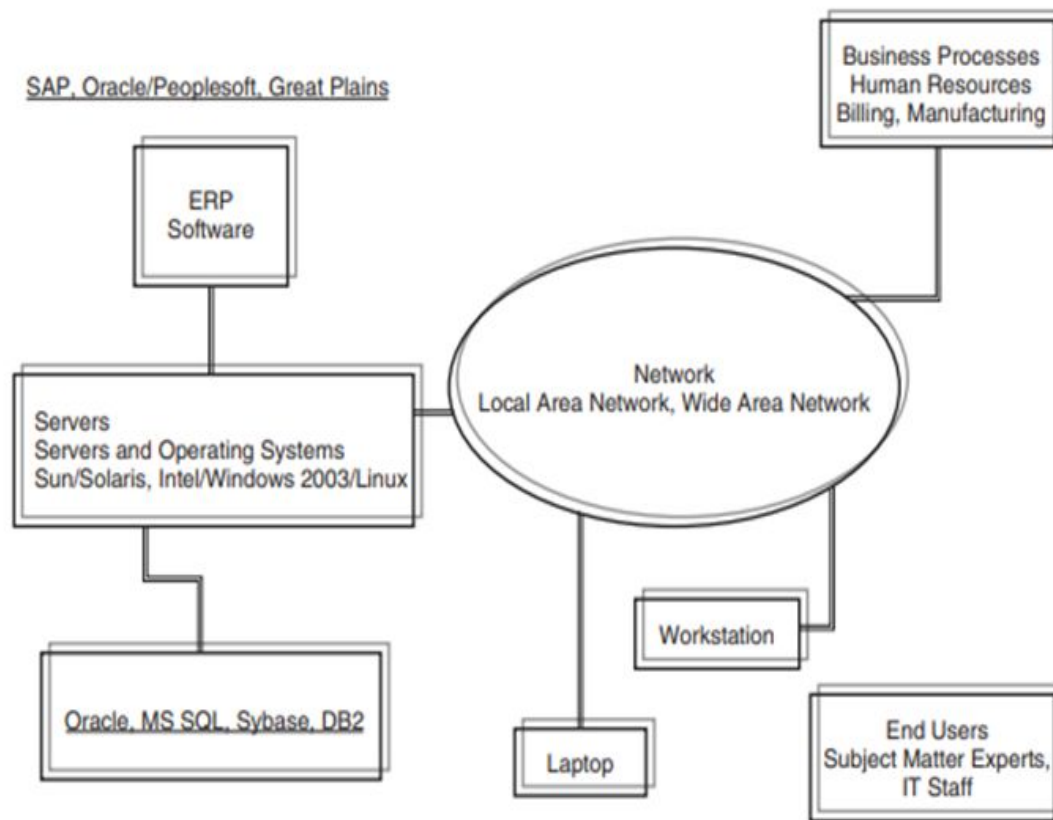


ERP System Components

- The key components for an ERP implementation are hardware, Software, database, processes, and people. These components must work together seamlessly for the implementation to be successful. The implementation team must carefully evaluate each component in relation to the others while developing an implementation plan.



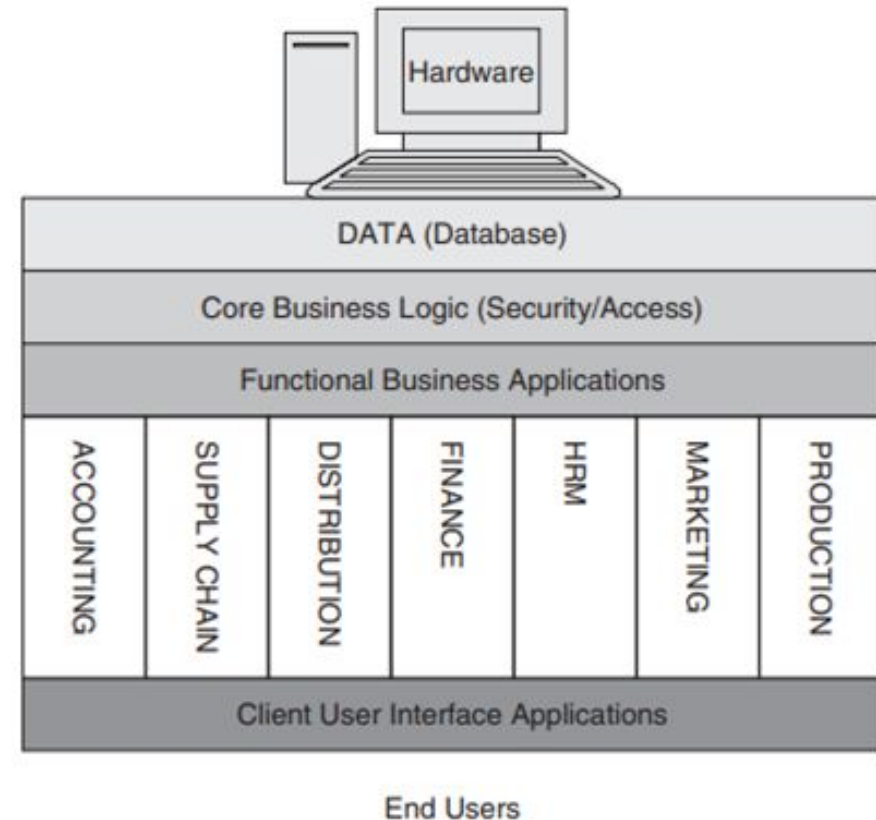
ERP Architecture



- A system's architecture is a blueprint of the actual ERP system and transforms the high-level ERP implementation strategy into an information flow with interrelationships in the organization. The process of designing ERP system architecture is slightly different from other IT architectures. Other IT architectures are driven by organizational strategy and business processes if the ERP vendor often drives purchased ERP architecture.

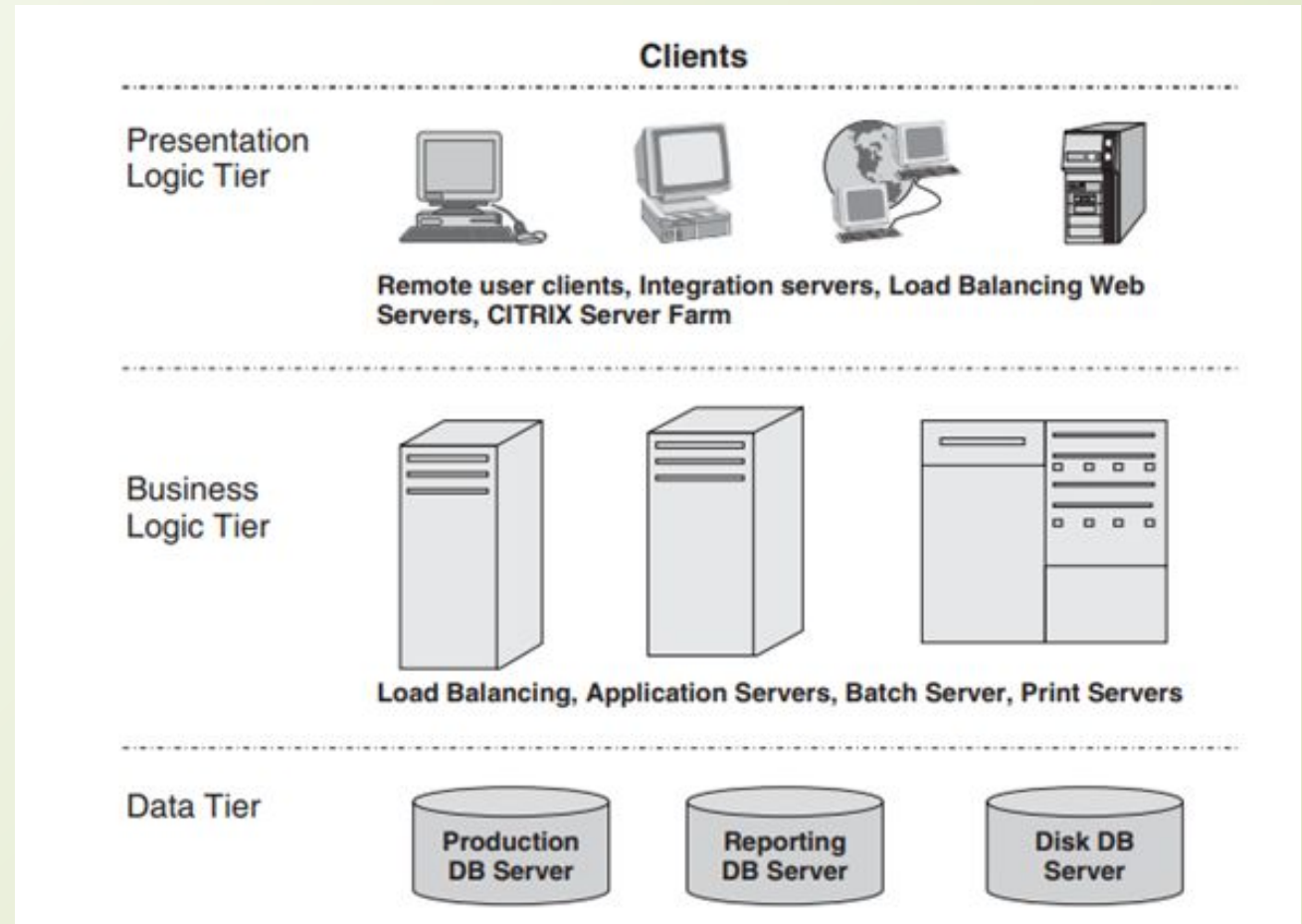
ERP Architecture(conti)

- An ERP package can have a very different implementation outcome from one organization to another. In the architecture of a large university, an ERP system can be very complex and must be designed and tested thoroughly before implementing it in the organization.



ERP Architecture(conti)

- The logical architecture, shown in Figure 1-6, focuses on supporting the requirements of the end-users, whereas the physical architecture focuses on the efficiency (cost, response time, etc.) of the system.





Benefits and Limitation of ERP

Integration of data and applications across functional areas of the organization. This means data can be entered once and used by all applications in the organization, improving the accuracy and quality of the data.

Consistency of the user interface across various applications means less employee training, better productivity, and cross-functional job movements.

Maintenance and support of the system improve as the IT staff is centralized and trained to support users' needs across the organization.

Data security and applications are enhanced due to better controls and centralization of hardware, Software, and network facilities.

Data conversion and transformation from an old to a new system can be extremely tedious and complex.

Consolidation of IT hardware, Software, and people resources can be cumbersome and difficult to attain.

Retraining IT staff and personnel to the new ERP system can produce resistance and reduce productivity over time.

The complexity of installing, configuring, and maintaining the system increases, requiring specialized IT staff, hardware, network, and software resources.



Choose your Team Mate!

